



InClass Assignment 5

Component Communication (@input / Component Splitting)

Objective: Master component decomposition by spinning up a child component and passing data downward using the @input decorator.

Part 1: Generate the Child Card Component

Open your secondary terminal and generate a new standalone component to represent an individual item card:

ng generate component product-card

Part 2: Configure the Input Decorator

Open `src/app/product-card/product-card.ts`. Import `Input` from `@angular/core`. Set up an input property to receive a product object:

```
import { Component, Input } from '@angular/core';

@Component({
  selector: 'app-product-card',
  standalone: true,
  imports: [],
  templateUrl: './product-card.html',
  styleUrls: ['./product-card.css'] // or template fallback style
})
export class ProductCardComponent {
  @Input() productData: any;
}
```

Part 3: Move Card Template Layout

Open `src/app/product-card/product-card.html` and move the card layout structure here:

```
<div style="border: 1px solid #eee; padding: 15px; border-radius: 5px;
margin-bottom: 10px;"
  [style.opacity]="productData.inStock ? '1' : '0.5'">
  <h3>{{ productData.name }}</h3>
  <p>Price: ${{ productData.price }}</p>

  @if (!productData.inStock) {
    <span style="color: red; font-weight: bold;">Out of Stock</span>
  }
</div>
```

Part 4: Wiring the Child Component into the List

1. Open `src/app/product-list/product-list.ts`. Import `ProductCardComponent` and add it to the component's imports: `[...]` array.
2. Open `src/app/product-list/product-list.html`. Replace the raw inner template block with the new custom selector, passing the item data through property binding:

```
<div style="padding: 20px; border: 1px solid #ddd; margin-top: 15px;">
  <h2>Available Items</h2>

  @for (product of products; track product.id) {
    <app-product-card [productData]="product"></app-product-card>
  }
</div>
```

Part 5: Submission

Submit all modified files across the `product-list` and `product-card` directories.



100 % 1 of 1 topics complete

In Class Assignment 5



Assignment